
Algorithm 1 Rule-based control loop

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1: Read pH, temperature, level, load-cell values, LDR values and dashboard
   commands.
2: if manual override is active then
3:   Execute authorized user command.
4:   Log the action.
5: end if
6: for each container  $i$  do
7:   Check the scheduled trigger  $T_i$ .
8:   Reserve  $W_i(t)$ .
9: end for
10: if dispensing is due then
11:   Verify the expected material.
12:   if verification fails then
13:     Block dispensing.
14:     Alert the user.
15:   else
16:     Open the servo lid.
17:     while dispensing do
18:       Continuously compare mass reduction with  $D_i$ .
19:       if target dose reached then
20:         break
21:       end if
22:       if timeout or fault condition then
23:         break
24:       end if
25:     end while
26:     Close the lid.
27:   end if
28: end if
29: if  $\text{pH}(t) < 6.5$  then
30:   Execute the lime-correction rule with feedback stopping.
31: end if
32: if level or temperature thresholds are violated then
33:   Operate the relevant pump until setpoint or timeout condition is met.
34: end if
35: Update Blynk dashboard values, actuator states, alerts and event logs.
36: Apply solar-tracker dead-band rule.
37: Return to the monitoring state.
```
